

MOHAMED MENSI

Network Security & Cybersecurity Engineer

Tunis, Tunisia | +216 51 306 136 | mohamed.mensi24@gmail.com

LinkedIn : [linkedin.com/in/mohamed-mensi/](https://www.linkedin.com/in/mohamed-mensi/) | Github : github.com/mohamed-mensi | Portfolio : mohamed-mensi.github.io

PROFESSIONAL SUMMARY

Cybersecurity and network security engineer (M1, ISGT) with hands-on experience in enterprise network infrastructure, penetration testing, and cloud security. Designed and validated a full VPN-MPLS backbone with OSPF/BGP, VRF-Lite, HSRP, and EtherChannel in GNS3. Designed and deployed a multi-stage attack chain exploiting CVE-2025-1094 (PostgreSQL UTF-8 encoding bypass) with JWT forgery and RCE for an OffSec UGC lab machine submission. Delivered 20+ security workshops to 50+ students as GDSC Cybersecurity Lead. Currently pursuing ISO 27001 knowledge in the MSSSI program. Seeking SOC analyst, network security, or junior penetration testing internships/positions.

TECHNICAL SKILLS

Network Security: Cisco IOS, OSPF, BGP, MPLS/VPN, VRF, VLAN, HSRP, EtherChannel (LACP), STP, GNS3, Packet Tracer, Wireshark

Offensive Security: Penetration Testing, Vulnerability Assessment, Burp Suite, Nmap, OWASP Top 10, CVE exploitation, CTF (forensics, web, network)

Cloud & Infra: OpenStack, Huawei ManageOne, VM lifecycle automation, Docker, Linux (Ubuntu/Kali), REST APIs

Programming: Python (scripting, automation), Java, Spring Boot, Flask, Bash

PROFESSIONAL EXPERIENCE

Cloud Resource Reservation & Billing System (PFE Internship)

CCK (El Khawarizmi Computing Center) | Feb – Jun 2025

Huawei Partner Project | Supervised by Dr. Amine Kechiche & Mr. Badis Saadi

Feb 2025 – May 2025

- Designed and deployed a multi-tenant cloud resource management platform for academic institutions, enabling secure ECS provisioning, reservation, and billing across three isolated university tenants using OpenStack REST APIs and Huawei ManageOne.
- Implemented role-based access control (RBAC) and tenant isolation policies in a Spring Boot backend, enforcing least-privilege access and preventing cross-tenant resource exposure in shared cloud infrastructure.
- Developed a Flask-based API orchestration layer acting as a secure abstraction over Huawei ManageOne, reducing direct exposure to cloud APIs and improving system control, security boundaries, and maintainability.
- Integrated Huawei IAM-based authentication, ECS lifecycle management (create/start/stop), real-time resource monitoring, and scheduling logic with availability validation for reservation workflows.
- Built hybrid billing system (fixed + pay-as-you-go), automated invoice generation, and resource usage tracking across tenants.
- Delivered under Agile methodology (Scrum, 4 sprints) with full UML design documentation and formal academic reporting.

github.com/mohamed-mensi/cloud-reservation-billing

IT Infrastructure Intern | Astral, Ben Arous

Aug 2023 – Sep 2023

- Assessed and reconfigured internal switch topology and server connectivity, identifying network bottlenecks and documenting infrastructure layout for IT continuity and security baselining
 - Migrated manual inventory to a digital tracking system, improving data integrity and auditability
-

SECURITY & NETWORK INFRASTRUCTURE PROJECTS

VPN-MPLS Enterprise Backbone — Network Lab | ISGT | GNS3

2025 – 2026

- Engineered a full VPN-MPLS backbone across 4 Cisco c7200 PE/P routers: configured OSPF as the IGP, iBGP for VPN prefix exchange, and VRF-Lite with RD/RT 100:3 for L3VPN tenant isolation — replicating enterprise ISP segmentation architecture
- Configured distribution-layer redundancy: HSRP active/standby gateway failover and EtherChannel (LACP) port-channel aggregation across IOU-L2 switches, with per-VLAN STP root bridge election for VLANs 10, 15, 20, 300
- Validated inter-VLAN routing, DHCP relay, and OSPF route redistribution from access layer to MPLS core; confirmed CE-to-CE reachability across L3VPN tunnel
- Systematically diagnosed and resolved 7 infrastructure issues: IOU license hostname validation failure, GNS3 VM VT-x hypervisor crash under Windows Hyper-V, EtherChannel LACP port suspension from misconfigured STP priorities, and unicast frame drops at the Dynamips/GNS3-VM boundary

github.com/mohamed-mensi/vpn-mpls-enterprise-network

Offensive Security Lab Engineering — “Vitals” UGC Machine | OffSec Proving Grounds Candidate

2026

- Designed and developed a realistic vulnerable healthcare infrastructure VM (“Vitals”) for submission to the OffSec Proving Grounds UGC program, simulating a misconfigured hospital patient monitoring environment
- Built a multi-stage attack chain leveraging CVE-2025-1094 UTF-8 encoding bypass against a Flask/PostgreSQL REST API to extract JWT secrets and escalate privileges through forged administrative access
- Implemented privilege escalation path via insecure PostgreSQL archive_command execution and misconfigured systemd service permissions, leading to root compromise
- Engineered realistic enumeration paths, embedded attack-surface hints, Swagger API discovery, JWT abuse scenarios, and database-driven lateral movement techniques to emulate real-world penetration testing workflows
- Developed the infrastructure using Ubuntu Server, Flask, PostgreSQL, Linux service hardening misconfigurations, and REST API authentication logic

github.com/mohamed-mensi/vitals-offsec-lab

AI-Driven SOC Analyst Assistant | Spring Boot, React, FastAPI, ML

2025

- Developed an ML-assisted SOC alert prioritization platform combining attack classification and incident severity prediction for high-volume security operations environments
- Engineered a dual-model classification pipeline using Random Forest models trained on the CIC-IDS2017 dataset to classify 11 attack categories and predict 4 alert priority levels
- Built a modular architecture integrating FastAPI inference services, Spring Boot backend orchestration, and React-based analyst dashboards with Swagger-documented REST APIs
- Implemented feature engineering and preprocessing workflows across 18 network flow features, handling severe dataset imbalance through weighted classification strategies
- Analyzed operational tradeoffs including inference latency, false positives/negatives, batch vs real-time processing, and scalability constraints in SOC environments

github.com/mohamed-mensi/soc-ml-service

LEADERSHIP & SECURITY COMMUNITY

- Technical Lead** | Securinets ISGT (University Cybersecurity Club) Nov 2025 – Present
- Lead offensive and defensive security curriculum, coordinating CTF training, tool workshops (Wireshark, Burp Suite, Nmap), and hands-on network security labs for club members
 - Mentor junior members on network protocol analysis, vulnerability scanning, and basic penetration testing methodology
- Cybersecurity Lead** | Google Developer Student Club (GDSC) Nov 2022 – May 2024
- Designed and delivered 20+ penetration testing and network security workshops (50+ students), covering web application attacks (OWASP Top 10), traffic analysis with Wireshark, and network reconnaissance with Nmap
 - Built the cybersecurity training curriculum from scratch, establishing the club's technical security practice

EDUCATION

- Master in Information Systems Security (M1)** | ISGT — Institut Supérieur de Gestion de Tunis Sep 2025 – Present
- Core: Network Security, Linux Administration, ISO 27001, Django, Cryptography, Risk Management
- Bachelor in Computer Science** | ISAMM — Institut Supérieur des Arts Multimédia de la Manouba 2022 – 2025
- Core: Java, Cloud Computing, Angular, Spring Boot, Flask, Software Engineering, Networking Fundamentals

CERTIFICATIONS & TRAINING

- Cisco CCNA1
- Cisco Linux Essentials
- Cisco CCNA2 (In Progress)
- ISC2 CC — Certified in Cybersecurity (In Progress)

LANGUAGES

Arabic (Native) | French (Fluent) | English (Professional Working Proficiency)